

Yishun JC H2 Chemistry 9647 Preliminary Examinations 2011

Suggested Answers

Paper 1: Multiple Choice Questions

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
D	B	C	A	B	C	B	D	A	D
Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
C	C	B	A	B	C	A	C	D	B
Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30
C	D	A	C	D	A	A	C	D	B
Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40
A	D	D	B	B	D	A	D	D	C

Paper 2: Structured Questions

- 1 (a) (To ensure that the total volume of the resultant solution remains the same across all experiments – already stated in the question)
This is so that the volume of each reactant is proportional to its concentration in the resultant solution.
(or when volume of that reactant added doubles, its concentration in the resultant solution doubles)

(b) $rate = \frac{1}{time}$ or $rate = \frac{k}{time}$ or $rate \propto \frac{1}{time}$

- (c) (i) rate of reaction $\propto [S_2O_3^{2-}]$ or the reaction is first with respect to $S_2O_3^{2-}$.

- (ii) $S_2O_3^{2-}$ is involved in the rate determining step of the mechanism
(or equivalent)

(d) (i)

experiment	volume of 2.00 mol dm ⁻³ Na ₂ S ₂ O ₃ / cm ³	volume of 0.010 mol dm ⁻³ NaAsO ₂ / cm ³	volume of 2.00 mol dm ⁻³ CH ₃ COOH / cm ³	volume of water / cm ³	time taken for the bright yellow ppt to appear / s
1	20	20	40	20	t ₁
2	20	10	40	30	t ₂

- o volume of AsO₂⁻ is changed, but volume of S₂O₃²⁻ and CH₃COOH remains the same
and
total volume of solution remains the same (by varying V_{water} appropriately)
- (ii) 1. Using a measuring cylinder, add 20 cm³ of NaAsO₂, 40 cm³ of CH₃COOH, and 20 cm³ of distilled water into a 250 cm³ conical flask.
2. Using a second measuring cylinder, measure 20 cm³ of Na₂S₂O₃.
3. Pour the contents into the conical flask, start the stopwatch, and give the conical flask one swirl to mix the contents.
4. Stop the stopwatch when a bright yellow precipitate is formed
o Mentions the use of measuring cylinder or burette to measure the correct volume of the solutions (as given in the table).
and

the solutions must be mixed swiftly (added from beaker or measuring cylinder and **not** directly from burette)

- o $S_2O_3^{2-}$ kept apart from H^+ until the stopwatch is started
(i.e. use different measuring cylinders)
(distilled water can be added to $S_2O_3^{2-}$ or to AsO_2^- or ethanoic acid)
- o Starts stopwatch when the last solution is added to the conical flask
and
Stops stopwatch when bright yellow precipitate appears
- o Mentions swirling of solution (after addition of the last solution) to ensure homogeneity.

(e) **Either (Mathematical Method)**

Let the rate equation be: $rate = k[S_2O_3^{2-}]^x[AsO_2^-]^y[H^+]^z$

taking rate of reaction 1 divided by that of reaction 2:

$$\frac{rate_1}{rate_2} = \left(\frac{[S_2O_3^{2-}]_1}{[S_2O_3^{2-}]_2} \right)^x \times \left(\frac{[AsO_2^-]_1}{[AsO_2^-]_2} \right)^y \times \left(\frac{[H^+]_1}{[H^+]_2} \right)^z$$

$$\Rightarrow \frac{\frac{1}{t_1}}{\frac{1}{t_2}} = \left(\frac{20}{10} \right)^y \Rightarrow \frac{t_2}{t_1} = \left(\frac{20}{10} \right)^y \Rightarrow y = \frac{\lg\left(\frac{t_2}{t_1}\right)}{\lg(2)}$$

Or (Inspection Method)

from (d)(i),

order	relating time ratio to rate ratio	relating volume ratio to rate ratio
0	if time taken for experiment 1 and 2 are the same, rate for both reactions are the same	$\Rightarrow$ halving the volume of AsO_2^- does not affect the rate of reaction
1	if time taken for experiment 2 is double that of experiment 1, rate for experiment 2 is half that of experiment 1	$\Rightarrow$ halving the volume of AsO_2^- caused the rate to be halved
2	if time taken for experiment 2 is quadruple that of experiment 1, rate for experiment 2 is one-quarter that of experiment 1	$\Rightarrow$ halving the volume of AsO_2^- caused the rate to be one-quarter

Or (V x t method)

from (d)(i),

order	determining factor
0	t is constant
1	$V_{AsO_2^-} \times t = \text{constant}$
2	$(V_{AsO_2^-})^2 \times t = \text{constant}$

- (f) Both sodium metaarsenite and arsenic(III) sulfide contains arsenic which is highly toxic / poisonous (or words to that effect ~ but not corrosive).

and any of the following:

- o wear gloves – to prevent direct contact with the chemicals
- o wear protective masks (especially during pouring of sodium metaarsenite and disposal of reaction mixture) – to avoid ingestion of the arsenic compounds
- o other reasonable answers

2 (a) (i)
$$n_{Cr_2O_7^{2-}} = \frac{1}{6} \times n_{Fe^{2+}} = \frac{1}{6} \times 1.00 \times 22.50 \times 10^{-3} = \frac{1}{6} \times 0.0225 = 3.75 \times 10^{-3} mol$$

(ii)
$$n_{Cr_2O_7^{2-}} (\text{in } 60cm^3) = \frac{3.75 \times 10^{-3}}{45} \times 60 = 0.0150 mol$$

$$n_{Cr_2O_7^{2-}} (\text{reacted}) = n_{Cr_2O_7^{2-}} (\text{reacted}) - 0.0150 mol = (0.500 \times 50.0 \times 10^{-3}) - 0.0150$$

$$= 0.0250 - 0.0150 = 0.0100 mol$$

(iii)
$$[C_2H_5OH] = \frac{n_{C_2H_5OH}}{V_{C_2H_5OH}} = \frac{\frac{3}{2} \times n_{Cr_2O_7^{2-}}}{10.0 \times 10^{-3}} = \frac{\frac{3}{2} \times 0.0100}{10.0 \times 10^{-3}} = \frac{0.0150}{10.0 \times 10^{-3}} = 1.50 mol dm^{-3}$$

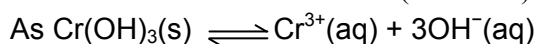
- (b) (i) negative

(ii)
$$n_{e^-} = \frac{Q}{F} = \frac{I \times t}{F} = \frac{10 \times 30 \times 60}{96500} = \frac{18000}{96500} = 0.187 mol$$

$$m_{Cr} = n_{Cr} \times M_{Cr} = \frac{1}{3} \times n_{e^-} \times M_{Cr} = \frac{1}{3} \times 0.187 \times 52.0 = 3.23 g$$

- (iii) The graphite electrode (anode) will be oxidized by the $O_2(g)$ produced to form $CO_2(g)$

(c) (i)
$$\text{solubility of } Cr(OH)_3 = \frac{1.27 \times 10^{-6}}{52.0 + 3(16.0 + 1.0)} = \frac{1.27 \times 10^{-6}}{103.0} = 1.23 \times 10^{-8} mol dm^{-3}$$



$$K_{sp} = [Cr^{3+}][OH^-]^3 = (1.23 \times 10^{-8}) \times (3 \times 1.23 \times 10^{-8})^3 = 6.24 \times 10^{-31} mol^4 dm^{-12}$$

- (ii) For precipitation, ionic product $> K_{sp}$

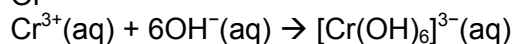
$$\Rightarrow (1.0 \times 10^{-5}) \times [OH^-]^3 > 6.24 \times 10^{-31}$$

$$\Rightarrow [OH^-] > 3.97 \times 10^{-9} mol dm^{-3}$$

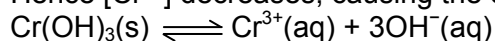
$$\Rightarrow n_{NaOH} = 3.97 \times 10^{-9} \times 100 \times 10^{-3} = 3.97 \times 10^{-10} mol$$

$$\Rightarrow m_{NaOH} = 3.97 \times 10^{-10} \times (23.0 + 16.0 + 1.0) = 3.97 \times 10^{-10} \times 40.0 = 1.59 \times 10^{-8} g$$

- (iii) When an excess of NaOH is added, the complex $[Cr(OH)_6]^{3-}$ is formed
 Or



Hence $[Cr^{3+}]$ decreases, causing the equilibrium:



shifts to the right,

$\Rightarrow Cr(OH)_3(s)$ dissolves

Or

Hence $[\text{Cr}^{3+}]$ decreases,

$\Rightarrow$ ionic product $< K_{\text{sp}}$ (formula may be given in part (i))

$\Rightarrow \text{Cr}(\text{OH})_3(\text{s})$ dissolves

3 (a) (i) $\text{Mg}^+(\text{g}) \rightarrow \text{Mg}^{2+}(\text{g}) + \text{e}$ (must have state symbols)

(ii) The successive removal of electrons is from:
an increasingly positive ion

Or

a species that has decreasing screening effect (but constant nuclear charge)

Or

a species where the distance between nucleus and outermost electron is decreasing

Hence the net attractive force between the outermost electron and the nucleus / effective nuclear charge increases

Or

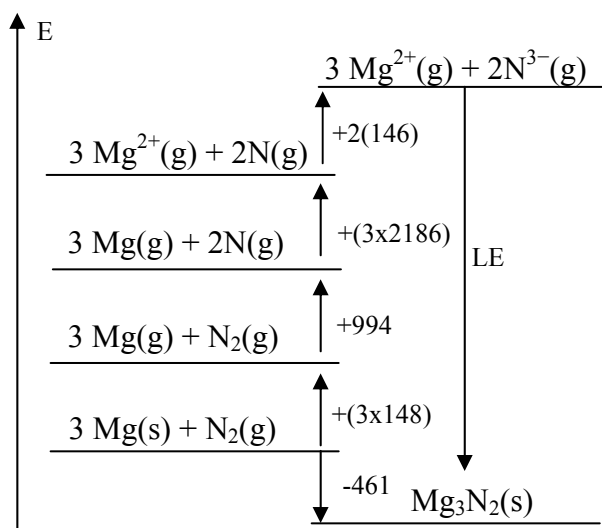
electrons are more tightly held by the nucleus
as successive electrons are removed.

(iii) For Mg, the third electron is removed from the $n = 2$ shell, while for Al it is the $n = 3$ shell $\Rightarrow$ Screening effect on the outermost electron on $\text{Mg}^{2+} \ll \text{Al}^{2+}$
(although nuclear charge of $\text{Mg} < \text{Al}$)

The net attractive force between the outermost electron and the nucleus / effective nuclear charge for $\text{Mg}^{2+} \gg \text{Al}^{3+}$

(b) (i) The lattice energy of an ionic compound is the energy evolved when 1 mol of a pure ionic solid is formed from its constituent gaseous ions.

(ii)



$$\Rightarrow LE = -461 - \{2(146) + 3(2186) + 994 + 3(148)\} = -8749 \text{ kJ mol}^{-1}$$

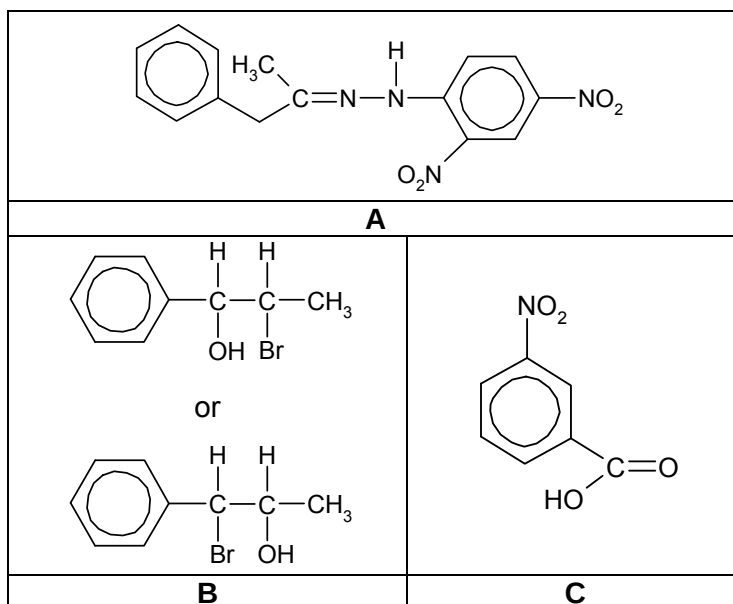
(iii) Although lattice energy of Mg_3N_2 more exothermic than MgO ,
The bond dissociation energy of the $\text{N} \equiv \text{N}$ bond is more endothermic than the $\text{O} = \text{O}$ bond

Or

The sum of the electron affinities of nitrogen is more endothermic than that of oxygen

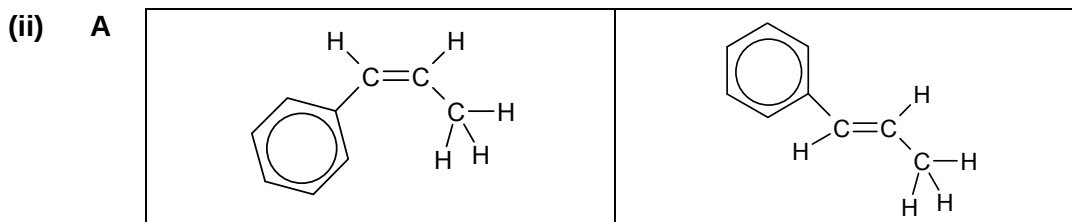
- (c) Mg reacts only with steam,
Ca, Sr, and Ba reacts with cold water with increasing vigour,
general equation: $M + 2H_2O \rightarrow M(OH)_2 + H_2$
- (d) (i) $Mg(NO_3)_2(s) \rightarrow MgO(s) + 2NO_2(g) + \frac{1}{2}O_2(g)$
- (ii) charge density / polarising power of $Mg^{2+} > Ba^{2+}$
(as ionic radius of $Mg^{2+} < Ba^{2+}$)
 $\Rightarrow NO_3^-$ is more significantly polarised for $Mg(NO_3)_2$,
 $\Rightarrow$ Ease of decomposition / weakening of the covalent bond in NO_3^- of $Mg(NO_3)_2 > Ba(NO_3)_2$
- (iii) ΔS is positive because of the increase in number of moles of gas
 $\Rightarrow$ increase in disorder / chaos / entropy
As $\Delta G = \Delta H - T\Delta S$,
When T increases, ΔG becomes more negative (because ΔS is positive)
 $\Rightarrow$ reaction becomes more feasible

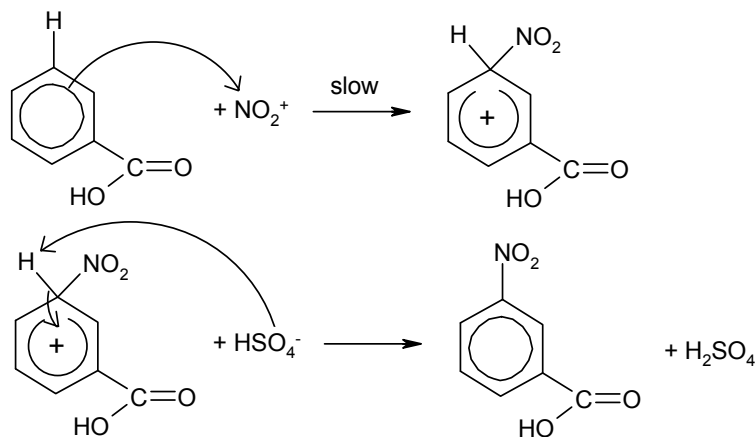
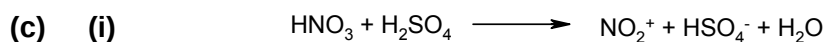
4 (a) (i)



Steps	Reagents and conditions
I	LiAlH ₄ in dry ether, or NaBH ₄ , or H ₂ (g) with Ni catalyst
II	concentrated H ₂ SO ₄ , 170 °C
III	KMnO ₄ (aq), H ₂ SO ₄ (aq), reflux

(b) (i) Elimination (not dehydration)

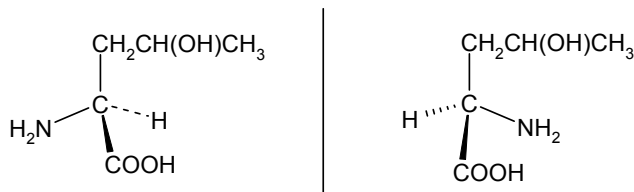
**B** 3-phenylpropene



- (ii) First order with respect to both $\text{C}_6\text{H}_5\text{COOH}$ and NO_2^+
As concentrated H_2SO_4 increases $[\text{NO}_2^+]$, it would lead to an increase in rate
- (iii) COOH is a ring de-activating group

- 5 (a) (i) It is a compound which has both the carboxylic acid group and the amine group attached to the same carbon atom.
Or
It is a compound with the following general structure: $\text{H}_2\text{N}-\text{CH}(\text{R})-\text{COOH}$

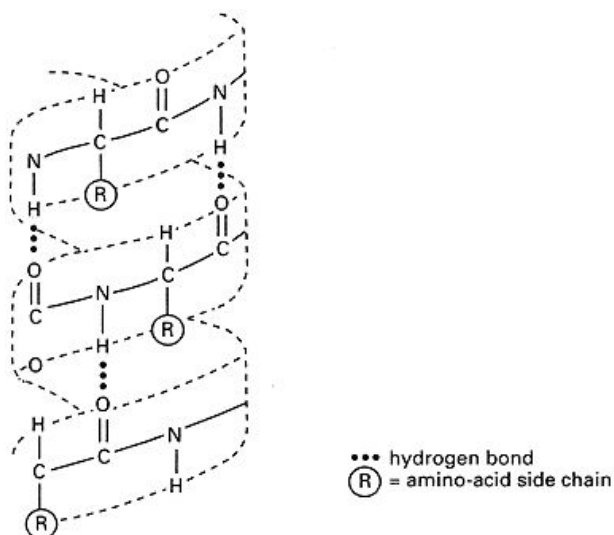
- (ii) Type of isomerism: Optical isomerism
Structures:



(iii)	with sodium carbonate	with bromoethane in ethanol
	$\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{C}-\text{COO}^- \text{Na}^+ \\ \\ \text{CH}_2 \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{CH}_3 \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{CH}_3\text{CH}_2-\text{N}-\text{C}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{CH}_3 \end{array}$

- (b) (i) Denaturation is the loss of activity caused to a protein when the interactions holding the secondary, tertiary or quaternary structures are broken. Heavy metal ions react with the disulfide bonds and disrupt the disulfide bridges causing the two chains to separate.

(ii)

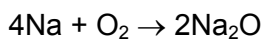


(c)

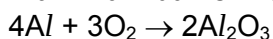
van der Waals' forces	ionic interactions

Paper 3: Free Response Questions

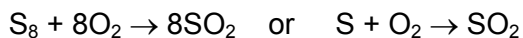
- 1 (a) (i) Sodium burns in excess air with an orange / yellow flame and forms a white solid.



Aluminium burns in excess air with a white flame and forms a white solid.



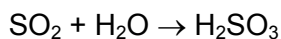
Sulfur burns in excess air with a blue flame and forms a colourless gas.



- (ii) $\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$

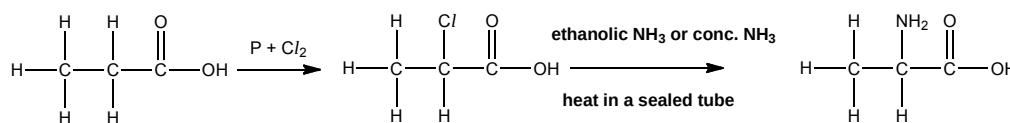
The alkaline solution formed gives blue colour with Universal Indicator.

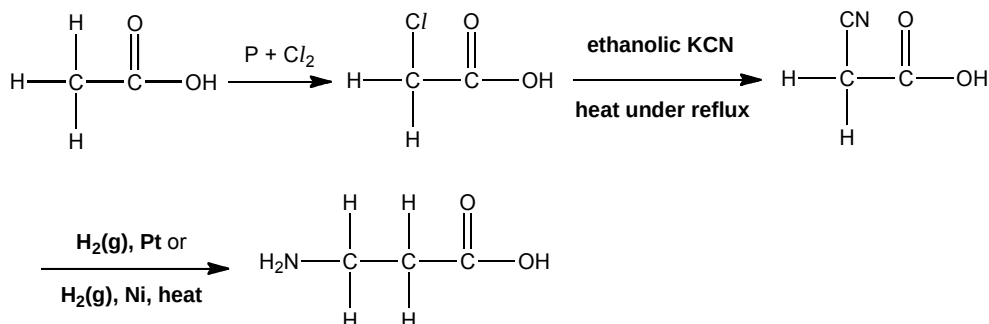
Al_2O_3 does not dissolve in water. The resulting mixture is neutral and gives green colour with Universal Indicator.



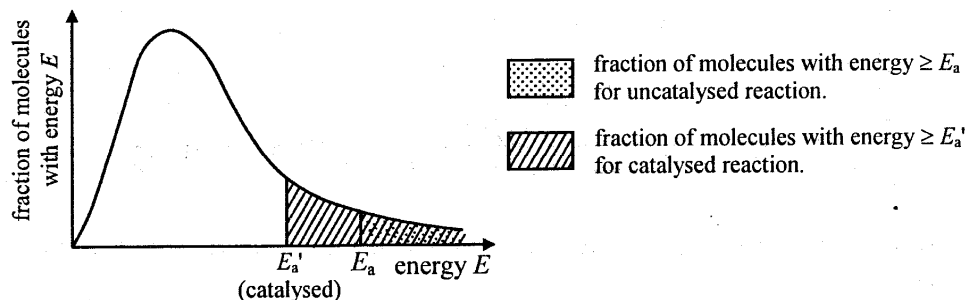
The acidic solution formed gives red colour with Universal Indicator.

- (b) (i) Synthesis of 2-aminopropanoic acid



Synthesis of 3-aminopropanoic acid

(ii)



A greater fraction of reactant particles have energy greater than or equal to the lowered activation energy in the catalysed reaction.
The frequency of effective collisions between the reactant particles increases.

(c) (i) HCl (ii) $2\text{SCl}_2(l) + 2\text{H}_2\text{O}(l) \rightarrow \text{SO}_2(\text{aq}) + \text{S}(s) + 4\text{HCl}(\text{aq})$ (iii) with $\text{AgNO}_3(\text{aq})$

white precipitate formed

 $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(s)$ and/or $\text{Ag}^+(\text{aq}) + \text{SO}_3^{2-}(\text{aq}) \rightarrow \text{Ag}_2\text{SO}_3(s)$ with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ orange $\text{K}_2\text{Cr}_2\text{O}_7$ turns green $\text{Cr}_2\text{O}_7^{2-} + 3\text{SO}_2 + 2\text{H}^+ \rightarrow 2\text{Cr}^{3+} + 3\text{SO}_4^{2-} + \text{H}_2\text{O}$

2 (a) (i) Boiling points of alkanes increase with the increasing number of carbon atoms due to greater number of electrons / larger electron cloud.

Thus, the electron cloud is more polarisable giving rise to stronger instantaneous-dipole induced-dipole / van der Waals forces of attractions.

Hence greater amount of energy is required to overcome the stronger instantaneous-dipole induced-dipole / van der Waals forces of attractions.

(ii) -42°C (around the same as C_3H_8)They have the same number of electrons / similar electron cloud size as C_3H_8 .

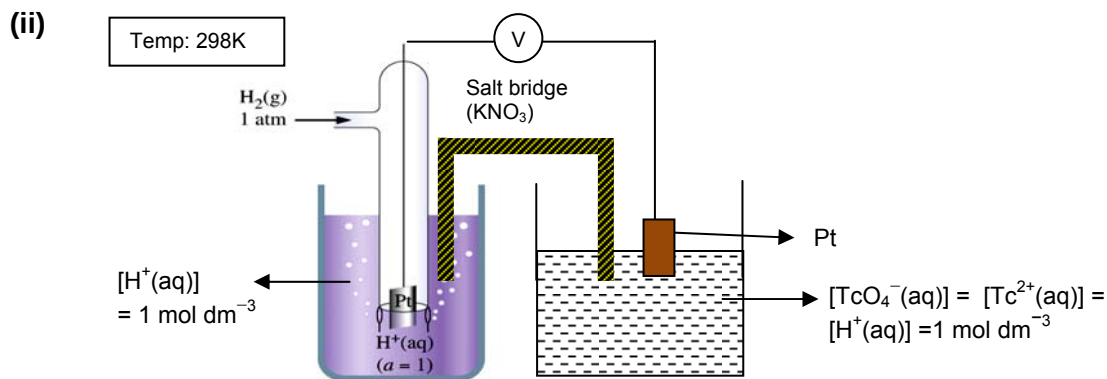
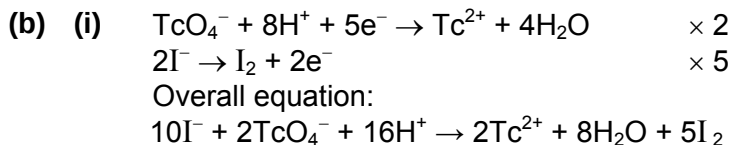
(iii) Alcohols exhibit intermolecular hydrogen bondings, which are stronger than instantaneous-dipole induced-dipole.

Hence greater amount of energy is required to overcome the hydrogen bonds in alcohols.

(iv) I methanol or ethanol

II If methanol (ethanol) had been present in the mixture it would have condensed as the first fraction at 65 °C (78 °C); or
The boiling temperature of methanol (ethanol) is below 97 °C. Hence the methanol (ethanol) could have boiled off before the first fraction is collected.

III Flammability of alcohol vapours or risk of fire or explosion



(c) Using the equation:

$$E = 0.32 + 0.029 \lg[I_2]$$

$$0.21 = 0.32 + 0.029 \lg[I_2]$$

$$\lg[I_2] = \frac{0.21 - 0.32}{0.029} = -3.79$$

$$\Rightarrow [I_2] = 10^{-3.79} = 1.61 \times 10^{-4} \text{ mol dm}^{-3}$$

$$\text{Amount of } \text{O}_3 = (1.6 \times 10^{-4}) \times (1.0 \times 10^{-2}) = 1.6 \times 10^{-6} \text{ mol}$$

Applying ideal gas law, $pV = nRT$,

$$V = \frac{(1.6 \times 10^{-6}) \times 8.31 \times 298}{0.24 \times (101 \times 10^3)}$$

$$= 1.66 \times 10^{-7} \text{ m}^3$$

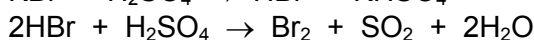
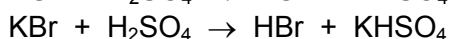
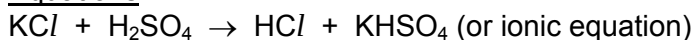
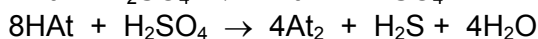
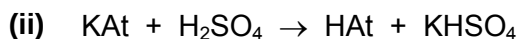
(d) (i) Amount of $\text{I}_2 = 0.300 \times (20.00 \times 10^{-3}) = 6.00 \times 10^{-3} \text{ mol}$
 Amount of ceronic acid = $0.328 \div 328 = 1.00 \times 10^{-3} \text{ mol}$
 $n(\text{I}_2) : n(\text{ceronic acid}) = 6 : 1$
 6 C=C double bonds

(ii) $\text{C}_{22}\text{H}_{32}\text{O}_2$ or $\text{C}_{21}\text{H}_{31}\text{COOH}$

3 (a) (i) Observations

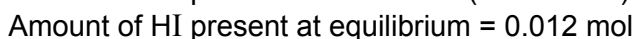
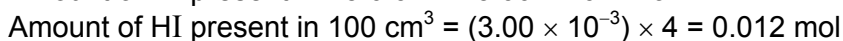
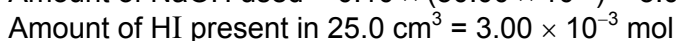
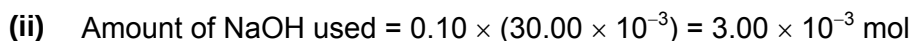
KCl: (steamy) white fumes

KBr: (steamy) white fumes and brown / orange fumes

EquationsExplanation:Cl⁻ / HCl is a weaker reducing agent than Br⁻ / HBr.

(As HAt is a stronger reducing agent than HI, it should reduce sulfur from the +6 oxidation state in H₂SO₄ to -2 in H₂S rather than 0 in elemental sulfur)

- (b) (i) The flask was rapidly cooled to prevent any disturbance to the equilibrium established at 400 °C so that the amount of H₂, I₂ and HI remained the same.



(iii)
$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$$

	H ₂ (g)	+	I ₂ (g)	$\rightleftharpoons$	2HI(g)
Initial / mol	0.01		0.01		0
Change / mol	- 0.006		- 0.006		+0.012
Equilibrium / mol	0.0004		0.0004		0.012

As the volume of the vessel is 2.0 dm³,

$$K_c = \frac{\left(\frac{0.012}{2.0}\right)^2}{\left(\frac{0.0004}{2.0}\right)\left(\frac{0.0004}{2.0}\right)} = 9$$

- (iv) At 600 °C, $K_c = 0.36$,

Let 2x be the number of moles of HI formed at equilibrium,

	H ₂ (g)	+	I ₂ (g)	$\rightleftharpoons$	2HI(g)
Initial / mol	0.01		0.01		0
Change / mol	-x		-x		+2x
Equilibrium / mol	0.01 - x		0.01 - x		2x

$$0.36 = \frac{\left(\frac{2x}{2.0}\right)^2}{\left(\frac{0.01-x}{2.0}\right)\left(\frac{0.01-x}{2.0}\right)}, \Rightarrow \sqrt{0.36} = \frac{2x}{0.01-x}, \Rightarrow 0.6(0.01-x) = 2x$$

$$\Rightarrow 0.006 - 0.6x = 2x, \Rightarrow 2.6x = 0.006, \Rightarrow x = 2.31 \times 10^{-3} \text{ mol},$$

$\Rightarrow$ Amount of HI at equilibrium = 4.62×10^{-3} mol

$\Rightarrow$ Amount of H_2 and I_2 at equilibrium = 7.69×10^{-3} mol

(v) Using the equation:

$$2.3 \lg \left[\frac{K_{c1}}{K_{c2}} \right] = \Delta H \left[\frac{1}{T_2} - \frac{1}{T_1} \right]$$

$$2.3 \lg \left(\frac{9}{0.36} \right) = \Delta H \left(\frac{1}{873} - \frac{1}{673} \right)$$

$$\Delta H = \frac{2.3 \lg \left(\frac{9}{0.36} \right)}{\left(\frac{1}{873} - \frac{1}{673} \right)} = -9460 \text{ J mol}^{-1}$$

$$\Rightarrow \Delta H = -9.46 \text{ kJ mol}^{-1}$$

(vi) $H_2(g) + I_2(s) \rightleftharpoons 2HI(g)$ $\Delta H_f^\circ = +54 \text{ kJ mol}^{-1}$

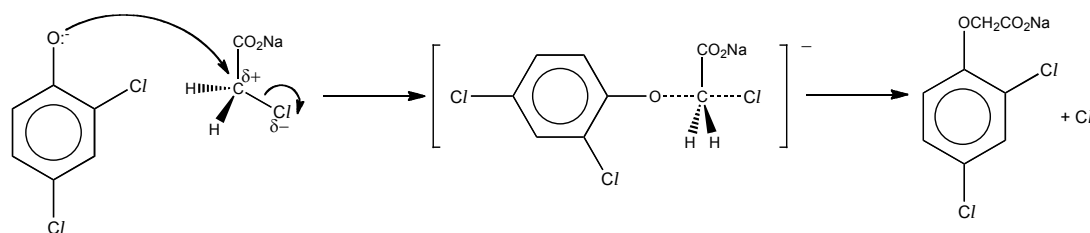
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ $\Delta H = -9.46 \text{ kJ mol}^{-1}$

At standard conditions, I_2 exists in the solid state.

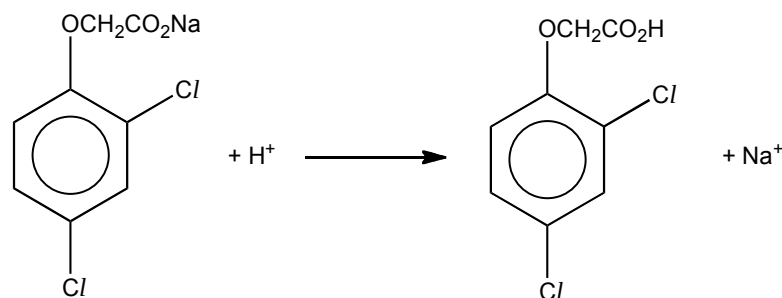
The difference in value is due to ΔH sublimation of I_2 .

(c) (i) The purpose of $OH^-(aq)$ is to generate phenoxide ion which is a stronger nucleophile than phenol for the nucleophilic substitution reaction.

(ii) S_N2 mechanism



(iii)



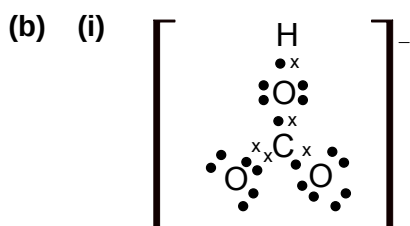
(iv)	test	2,4-D	2,4-dichlorophenol
	neutral FeCl_3 solution	no purple colouration	purple colouration
	or		
	$\text{Br}_2(\text{aq})$	no decolourisation	brown colour of $\text{Br}_2(\text{aq})$ decolourised
	and		
	$\text{PCl}_3/\text{PCl}_5/\text{SOCl}_2$	white fumes of HCl	no white fumes
	or		
	$\text{Na}_2\text{CO}_3 / \text{NaHCO}_3$	effervescence observed with CO_2 produced forming a white ppt in limewater	no effervescence

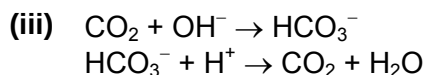
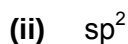
4 (a) structures of unknowns

P	Q
$\begin{array}{c} \text{O} \quad \text{OH} \\ \parallel \quad \\ \text{H}_3\text{C}-\text{C}-\text{CH}_2-\text{C}-\text{CH}_2\text{CH}_2-\text{COOH} \\ \\ \text{CH}_3 \end{array}$	$\begin{array}{c} \text{O} \quad \text{Cl} \quad \text{O} \\ \parallel \quad \quad \parallel \\ \text{H}_3\text{C}-\text{C}-\text{CH}_2-\text{C}-\text{CH}_2\text{CH}_2-\text{C}-\text{Cl} \\ \\ \text{CH}_3 \end{array}$
R	S
$\begin{array}{c} \text{O} \\ \parallel \\ \text{H}_3\text{C}-\text{C}-\text{CH}=\text{C}-\text{CH}_2\text{CH}_2-\text{COOH} \\ \\ \text{CH}_3 \end{array}$	
T	U
$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{OH} \end{array}$	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ \text{H}_3\text{C}-\text{C}-\text{CH}_2\text{CH}_2-\text{C}-\text{OH} \end{array}$

Explanations:

- **P** contains two $-\text{OH}$ groups that could undergo nucleophilic substitution with SOCl_2 .
- **P** contains ketone (and 3° alcohol) groups since it is not oxidised by acidified $\text{K}_2\text{Cr}_2\text{O}_7$.
- On heating with concentrated H_2SO_4 , **P** undergoes elimination to form alkene **R** and (intramolecular) esterification / condensation to form (cyclic) ester **S**.
- **R** (but not **S**) contains $\text{C}=\text{C}$ groups since it undergoes oxidative cleavage with KMnO_4 .
- **P**, **Q**, **R**, **S**, **T** and **U** contains the $\text{CH}_3\text{CO}-$ structure since they give positive triiodomethane test.
- **U** does not contain chiral carbon atom since it is not optically active.





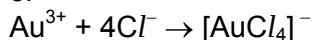
(iv)
$$pH = pK_a - \log \frac{[CO_2]}{[HCO_3^-]} = -\log(7.94 \times 10^{-7}) - \log\left(\frac{1}{20}\right) = 7.4$$

(c) (i)
$$[H^+] = \sqrt{c \times K_a} = \sqrt{0.10 \times 10^{-2.8}} = \sqrt{0.10 \times 1.58 \times 10^{-3}} = 1.258 \times 10^{-2} \text{ mol dm}^{-3}$$

$$pH = -\log(1.258 \times 10^{-2}) = 1.9$$

(ii) The strength of an acid depends on the stability of its conjugate base. The electron-withdrawing chlorine atom in acids **II** and **III** stabilises the conjugate bases by dispersing the negative charge on the O atom, hence making them stronger acids than **I**.For acid **III**, the chlorine atom is nearer to the $-CO_2^-$ group (than acid **II**) and disperses the negative charge on its conjugate base to a greater extent.
 $\Rightarrow$ Acid **III** is a stronger acid than acid **II**.5 (a) (i) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$
Copper can form a stable cation (Cu^{2+}) that has a partially filled 3d-subshell(ii) H_2O has 2 lone pairs of electrons and 2 bond pairs of electrons. However, when bonded to Cu^{2+} , it has only 1 lone pair of electrons and 3 bond pairs of electrons. When it is bonded to Cu^{2+} , there is no lone pair-lone pair repulsion and H_2O ligand only experiences lone pair-bond pair repulsion. Thus, the H-O-H bond angle is greater than 104.5° .

(b) (i) C: red / orange / yellow; D: blue / violet

(ii) C
Absorption is at lower wavelength/higher frequency.(c) The concentrated HNO_3 in *aqua regia* acts as an oxidising agent and oxidises Au to $Au(III)$ ion orThe aqueous Au^{3+} ion that is formed then reacts with concentrated HCl to form $[AuCl_4]^-$ or

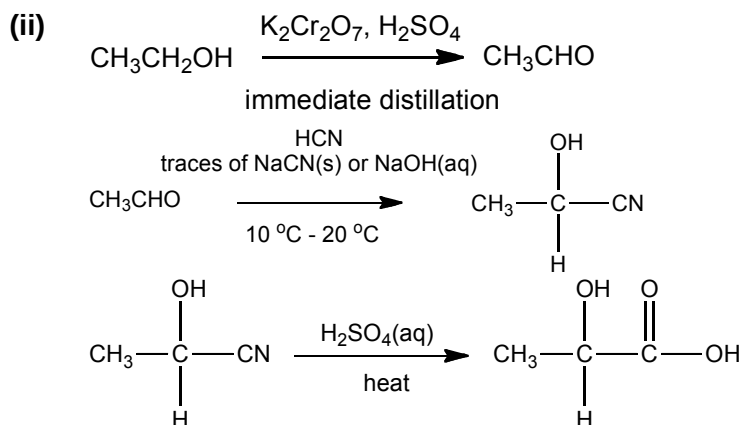
The formation of the complex shifts equilibrium (1) towards the right and hence gold eventually dissolves in the acid mixture.

(d) (i) structures of unknowns

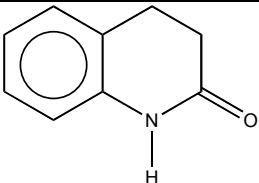
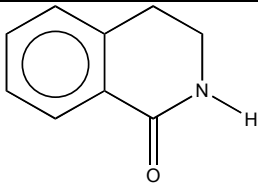
E	F	G
$ \begin{array}{c} Cl \quad O \\ \quad \\ CH_3 - C - C - Cl \\ \\ H \end{array} $	$ \begin{array}{c} Cl \quad O \\ \quad \\ CH_3 - C - C - OH \\ \\ H \end{array} $	$ \begin{array}{c} OH \quad O \\ \quad \\ CH_3 - C - C - OH \\ \\ H \end{array} $

Explanations:

- **E** hydrolyses in moist air to form white fumes of hydrogen chloride and reacts with water to form carboxylic acid. Thus **E** is an acid chloride.
- **F** has a chiral carbon atom since it is optically inactive.
- When reflux with NaOH(aq), **E** undergoes nucleophilic substitution reactions to form a hydroxycarboxylate anion, which upon acidification forms a hydroxycarboxylic acid.



(e)

Distinguishing Test		
warm with NaOH(aq) followed by addition of Br ₂ (aq)	brown colour of Br ₂ (aq) decolourised and appearance of a white ppt	no decolourisation and no ppt

~ END OF SUGGESTED ANSWERS ~